2025

A Decadal Analysis of Temperature Extremes and Rainfall Variability in Bikaner, India

REPORT SUBMITTED TO BCAP (BIKANER CLIMATE ACTION PLAN)

APARNA

TREND AND PATTERN OF HEAT WAVES IN BIKANER, RAJASTHAN (2014–2024)

Heat wave declaration criteria

Based on actual maximum temperature (for plains only)

- Heat Wave- When the actual maximum temperature is ≥45 °C
- Severe heat wave- When actual maximum temperature ≥47 °C

StudyArea - Bikaner, Rajasthan

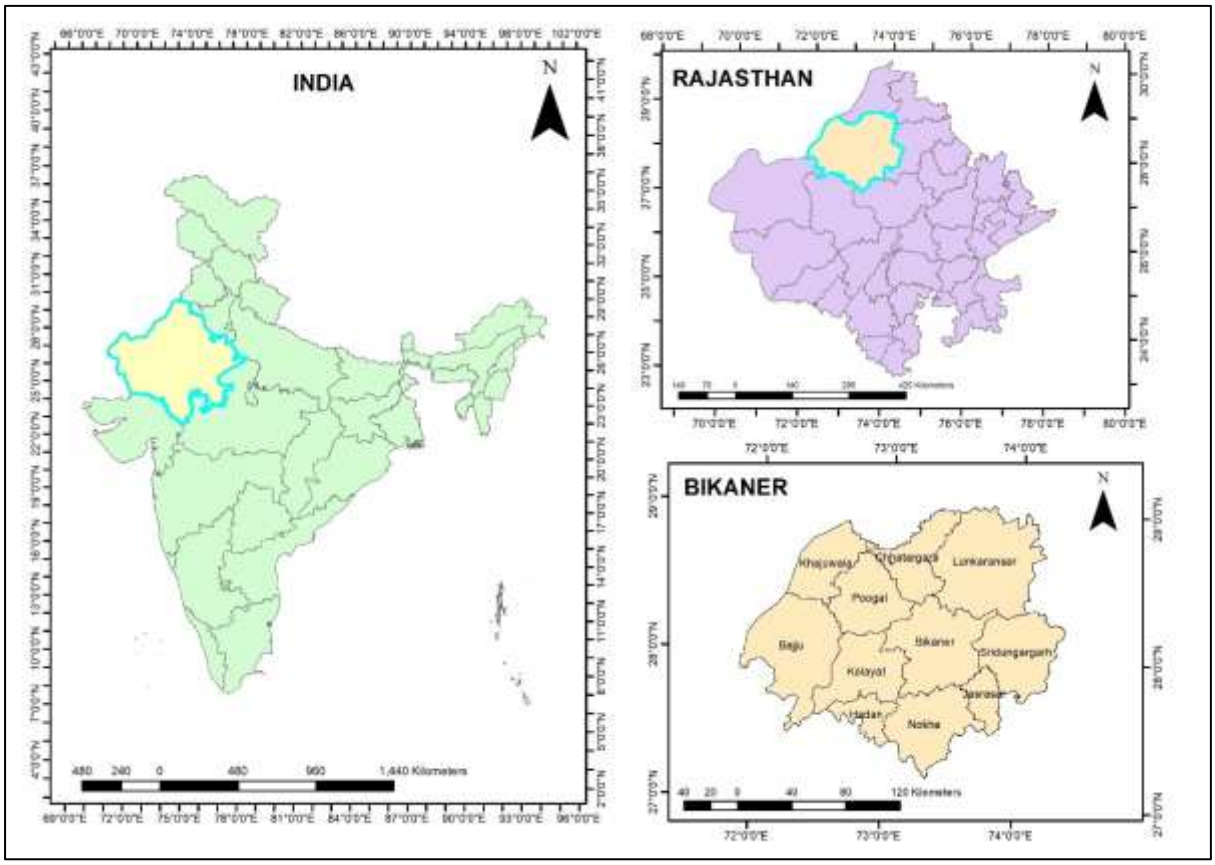


Figure 1 Study Area Map - Bikaner, Rajasthan

Data Source

Daily mean maximum temperature data is collected from NASA's Power Access Viewer for the period of 2014–2024 (March-September).

RESULTS

Extreme heat events

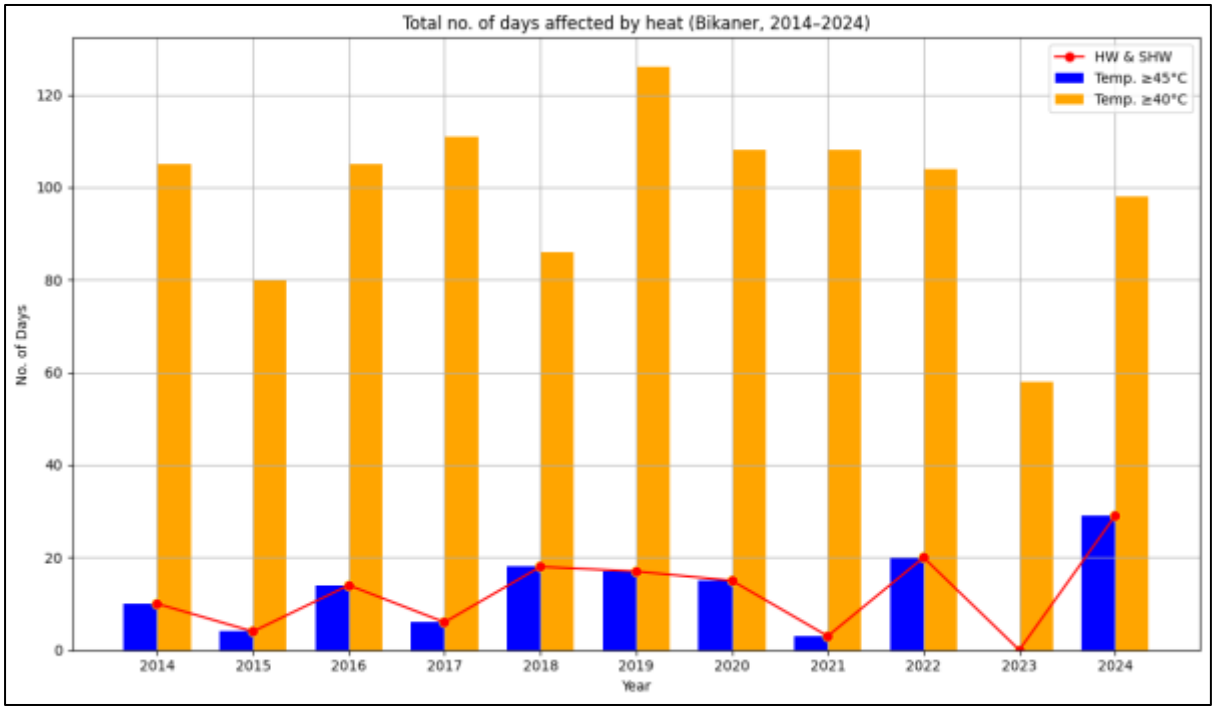


Figure 2 Total number of days with Heat Waves and Severe Heat Waves, and temperature is equal to or above 40 °C and 45 °C in a year from 2014–2024.

Key Observations:

- 2019 stands out with maximum days ≥40°C and high days ≥45°C, but a moderate HW/SHW count.
- 2024 has the highest number of HW/SHW days (30), also with a high count of days ≥45°C.
- 2023 shows an unusual drop in all categories, indicating a cooler or less extreme year.
- The general trend for days ≥40°C remains relatively high and consistent, suggesting persistent heat pressure over the decade.

Monthly Occurrence of Heat Wave Events

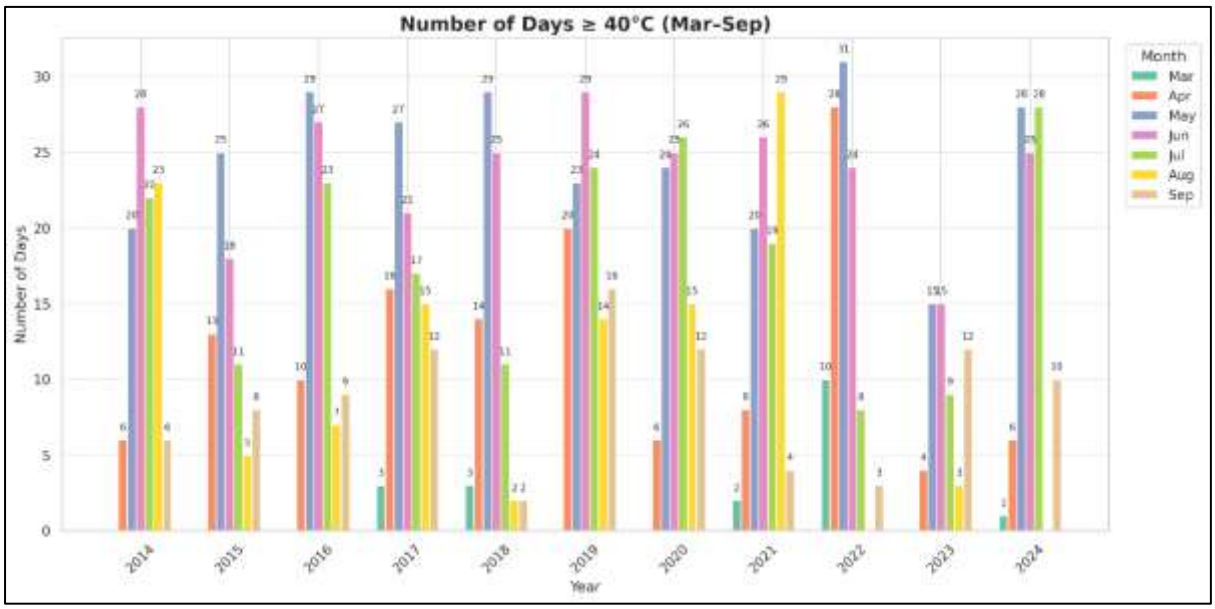


Figure 3 Number of days having temperature above 40 °C for the month of March-September (2014-2024)

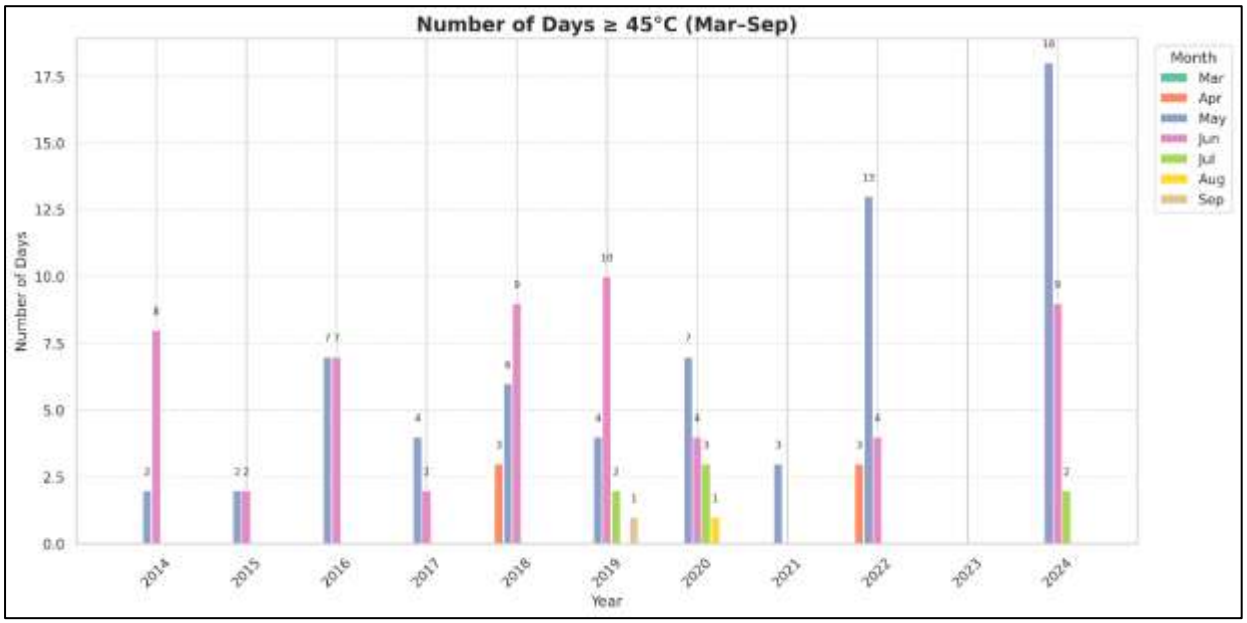


Figure 4. Number of days having temperature above 45 °C for the month of March-September (2014-2024)

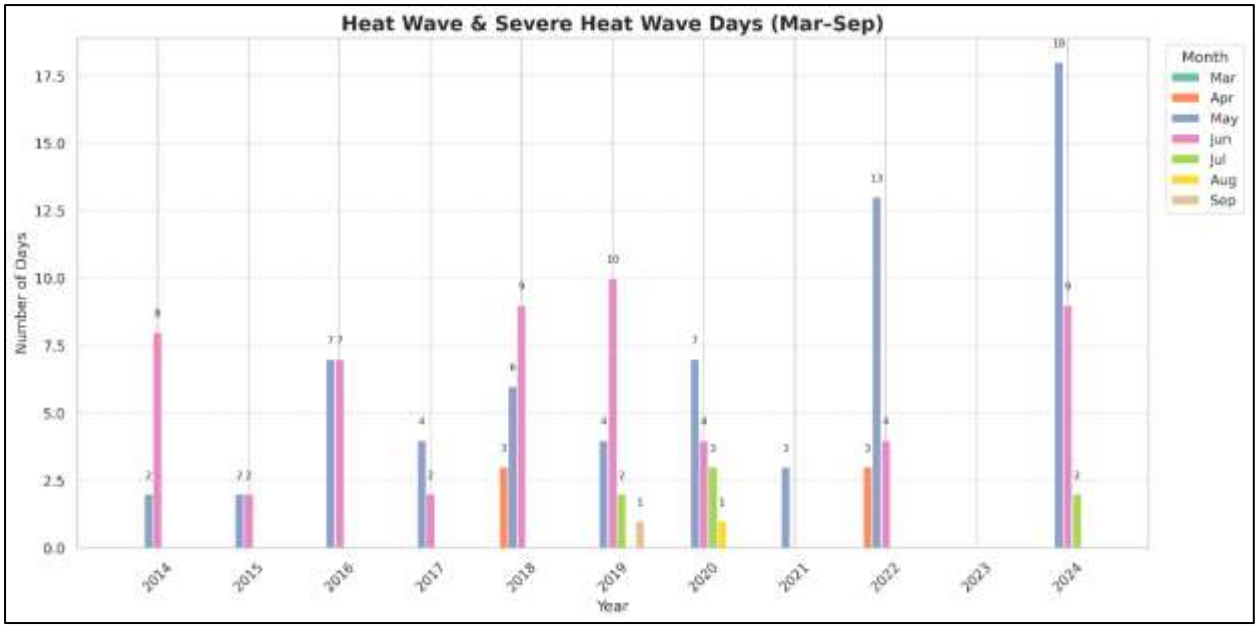


Figure 5 Number of days having HWs and SHWs for the month of March-September(2014–2024)

Key Observations:

- **Most Active Months for Heat Waves:**
  - May (blue) and June (pink) are the dominant months for HW/SHW events.
  - May is especially prominent in recent years (2022 and 2024) with 13 and 18 days, respectively.
  - June saw major activity in 2014, 2016, 2018, 2019, and 2024 (ranging 7–10 days).
- **Noteworthy Years:**
  - **2024:** Highest total HW/SHW days (30 days), mostly in May (18) and June (9), with some in July (2).
  - **2022:** Strong heat wave activity in May (13 days), plus contributions from April (3) and June (4).
  - **2019:** Distributed across months—June (10), May (4), July (2), August (1), September (1)—indicating an extended heat season.
  - 2018: April had 3 days; June (9) and May (6) were still dominant.
  - 2023: No HW/SHW days reported, in line with the previous graph.
- **Months Beyond May–June:**
  - **July**: Notable activity in 2019 and 2024.
  - **August & September:** Minor but unusual presence in 2019 and 2020, suggesting late-season heat waves.
  - **March:** Only 2020 and 2024 recorded heat wave activity.

Conclusion:

- The peak HW/SHW period is May–June, with May becoming more intense in recent years.
- Some years like 2019 and 2020 show heat waves spilling into monsoon months (July–September), which could point to shifting climatic patterns.
- 2024 stands out as the most heat-stressed year both in intensity and duration, suggesting a potentially worsening trend.

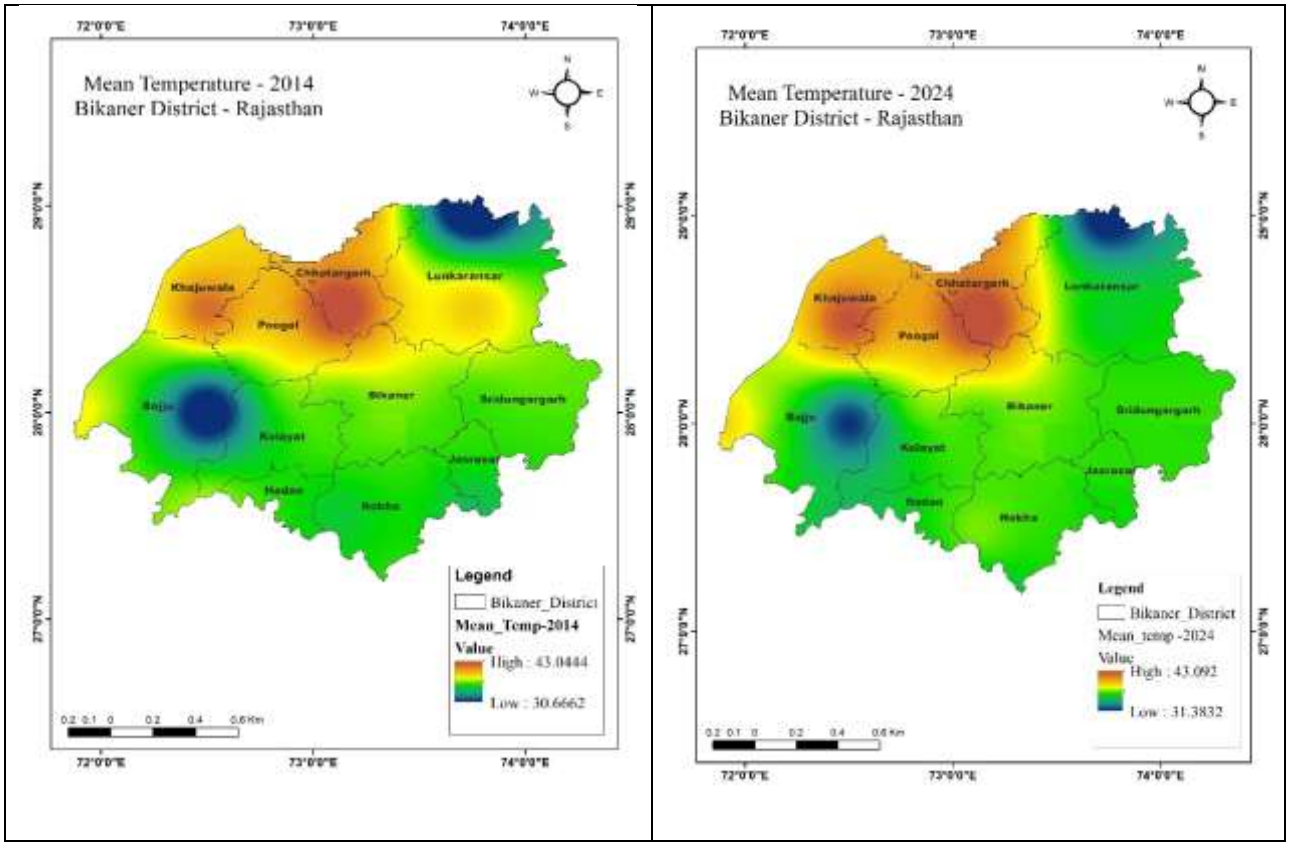


Figure 6 Monthly average maximum temperature of 2014 and 2024

The maps compare the monthly average maximum temperatures across the Bikaner district for the years 2014 (left) and 2024 (right). Here's a breakdown:

Observations by Comparison:

1. Overall Increase in Baseline Temperatures:

- The minimum average temperature increased from 30.67°C (2014) to 31.38°C (2024).
- This indicates a general warming trend even in the cooler areas of the district.

2. Hotspot Intensification and Spread:

- In 2014, the hottest zones (in orange/red) were localized mainly around Poogal and Chhatargarh.
- By 2024, these high-temperature zones have:
    - Expanded spatially (covering more of Chhatargarh, Khajuwala, and part of Lunkaransar).
    - Become more intense (consistent with the slight increase in the max temp value).

3. Cooling Zones Shrink:

- Cooler zones (blue- green) in 2014 are more extensive, especially in southern and central Bikaner.
- By 2024, these zones have shrunk, showing a reduction in climatic moderation, particularly in Kolayat, Nokha, and Sri Dungargarh.

4. Spatial Shifts:

- Some regions like Lunkaransar, which were moderately warm in 2014, appear hotter in 2024.
- Overall warming is more evenly distributed across the district in 2024 compared to 2014.

Average Maximum Temperatures:

Year	Min Avg Max Temp (°C)	Max Avg Max Temp (°C)
2014	30.67	43.04
2024	31.38	43.09
Change	+0.71°C (Min)	+0.05°C (Max)

- The minimum zone's temperature increased by ~0.71°C over 10 years, showing a clear warming of cooler zones.
- The maximum value barely changed, indicating the upper thermal ceiling is stable, but heat is becoming more spatially widespread.

Supporting Evidence from Heat Wave Graphs (Figure 5) :

- HW/SHW days increased dramatically, especially in 2022 and 2024.
- The number of days ≥45°C was much higher in recent years (2022, 2024).
- May and June are seeing more intense and prolonged heat.

Conclusion:

- While the hottest spots haven't gotten significantly hotter, the milder zones are warming quickly.
- This supports a ~1°C rise per decade in baseline max temperatures, especially in areas previously considered less heat-prone.
- This is consistent with regional and global climate trends, especially in arid and semi-arid zones like northwest Rajasthan.

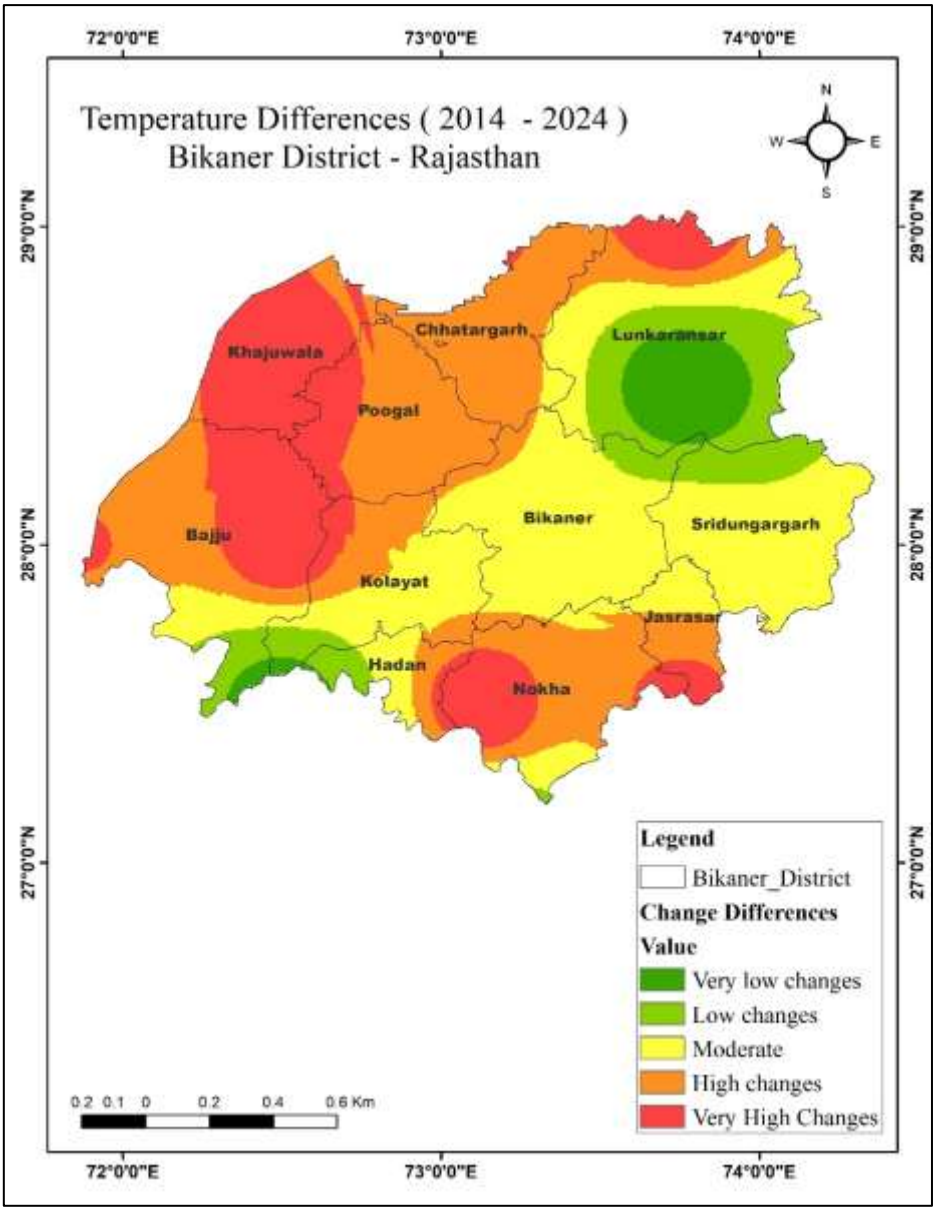


Figure 7 Change detection map of Monthly average maximum temperature (2014–2024)

This temperature difference map (2014–2024) for the Bikaner District vividly supports that certain areas have experienced nearly +1°C or more warming over the decade, aligning with the "1°C per decade" warming rate.

Change Class	Color	Interpretation
Very Low Change	Dark Green	Minimal or no change (Δ < ~0.2°C)
Low Change	Light Green	Mild warming (~0.2–0.4°C)
Moderate	Yellow	Moderate warming (~0.5–0.7°C)
High Change	Orange	Significant warming (~0.8–0.9°C)
Very High Change	Red	Extreme warming (Δ ≥ ~1°C) — aligns with global warming threshold concern

Spatial Patterns of Warming:

- Very High Changes are observed in:
    - Khajuwala, Nokha, Poogal, Chhattargarh, and pockets of Jasrasar.
    - These are arid interiors, vulnerable to desertification and solar heating.
- Moderate–High Changes dominate most of the district, including central Bikaner.
- Low or Very Low Changes appear around Lunkaransar, Hadan, and some southern fringes ,possibly due to:
    - Irrigation influence
    - Vegetation cover
    - Elevation or local microclimates

Summary:

- The red and orange zones confirm ~1°C increase, consistent with your "1°C/decade" hypothesis.
- Suggests localized mitigation/adaptation planning is needed, especially in hotspots like Nokha and Khajuwala.

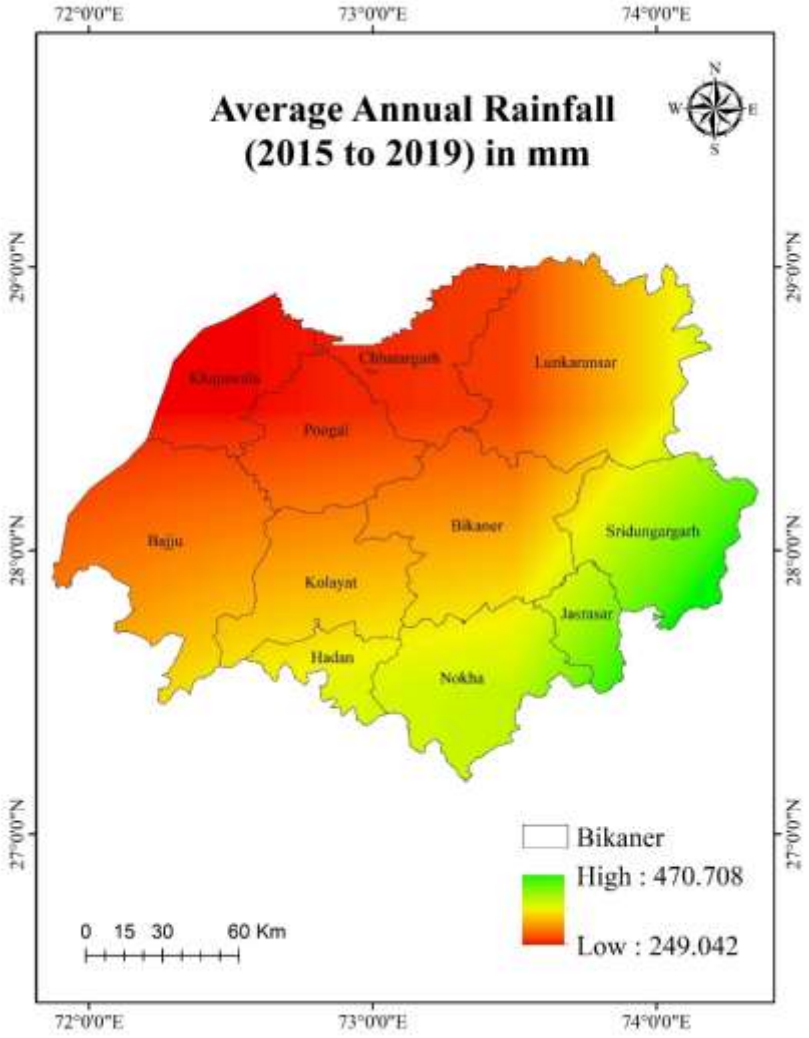
Conclusion

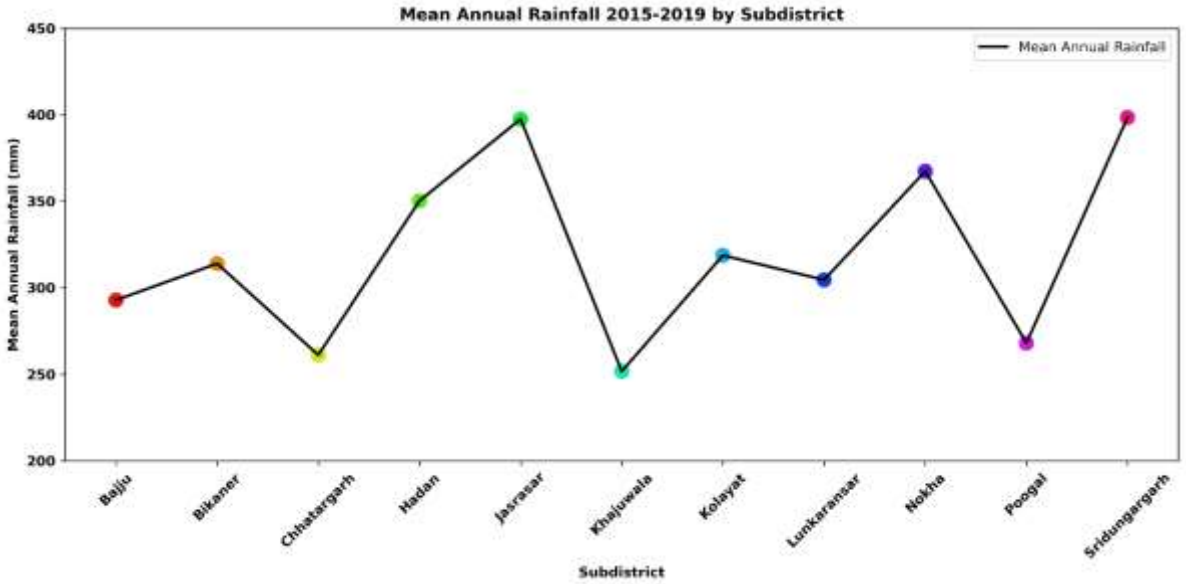
The study reveals a clear increasing trend in the duration, frequency, and intensity of heat waves in Bikaner District from 2014 to 2024. The summer season is advancing, with rising maximum temperatures observed as early as March, indicating that heat wave events are beginning earlier and persisting longer into the year. Analysis of temperature records and spatial data shows that the average daily maximum temperature has steadily increased across several months,especially from March to June with the year 2024 witnessing the highest number of extremely hot days (≥45°C). Subregions like Khajuwala, Nokha, Poogal, and Chhattargarh have experienced very high increases (≥1°C) in average maximum temperatures over the decade, marking them as critical heat hotspots. Meanwhile, central areas like Bikaner city show moderate to high warming trends.

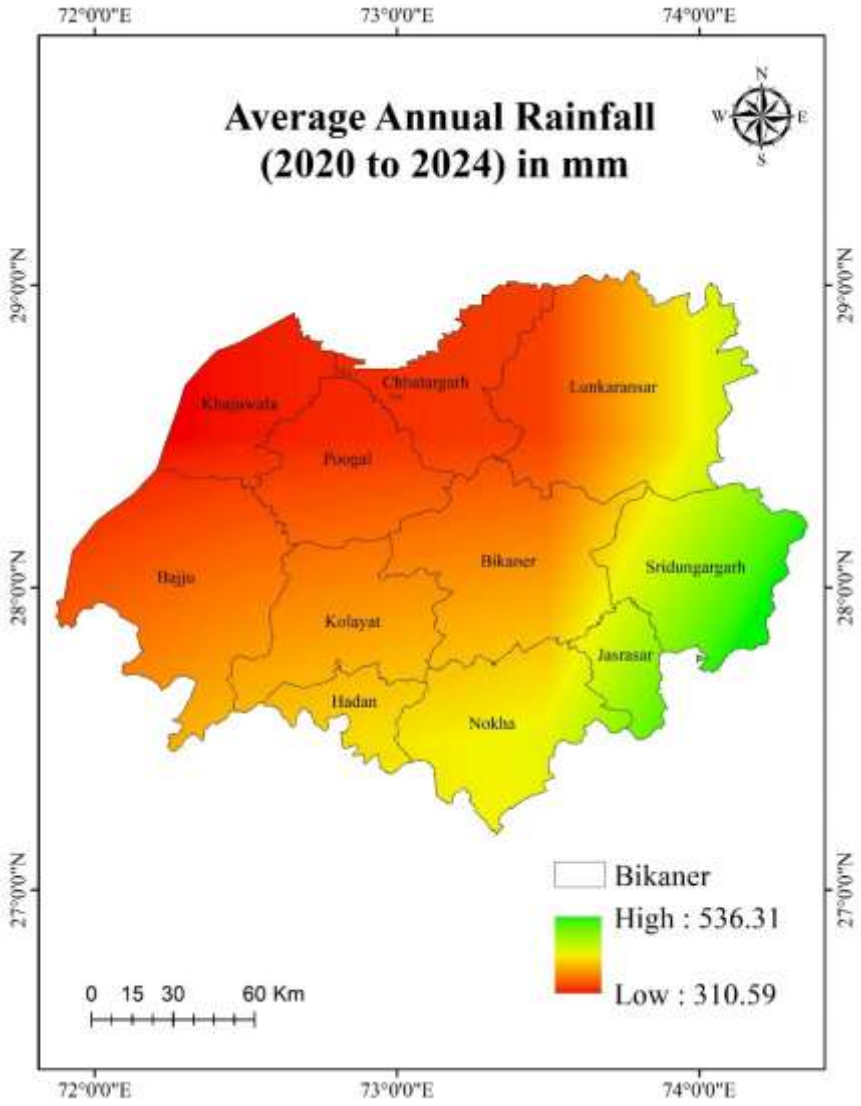
With a growing footprint of areas affected by severe heat and a rising number of extremely hot days, the population's vulnerability is expected to rise, especially among outdoor workers, children, and the elderly. Without targeted adaptation strategies, the risks to health, productivity, and livelihoods will continue to escalate. Therefore, region-specific heat action plans, early warning systems, community-level awareness campaigns, and the adoption of climate-resilient infrastructure are essential to mitigate the adverse effects of rising temperatures and frequent heat waves in Bikaner District.

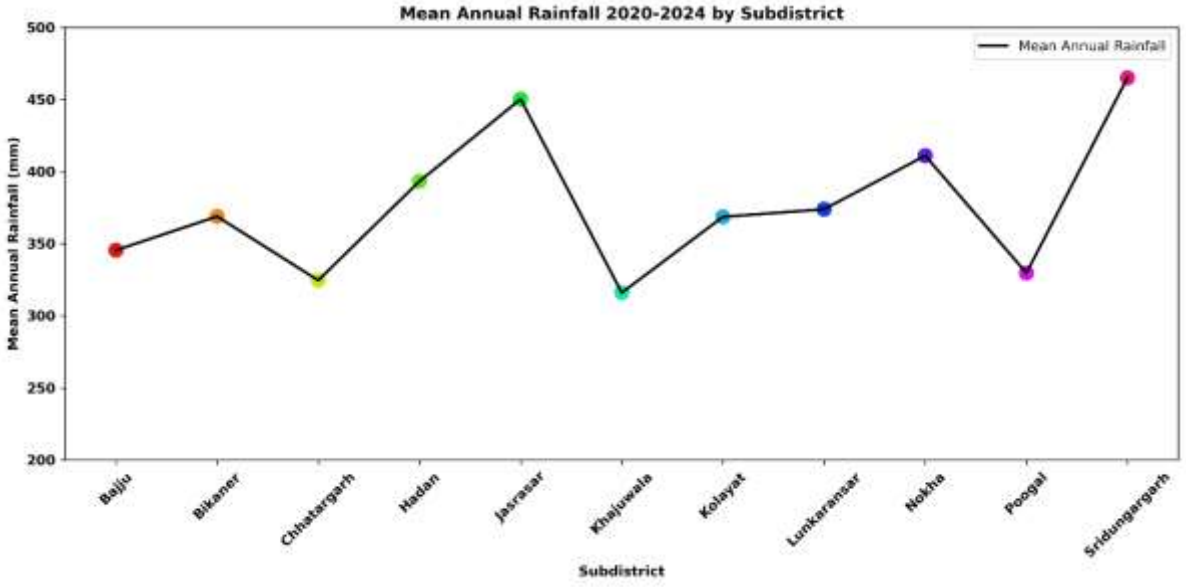
RAINFALL PATTERNS IN BIKANER OVER THE PAST DECADE

(2015-2024)






Average Annual Rainfall
(2020 to 2024) in mm
72°0'0"E
73°0'0"E
74°0'0"E
29°0'0"N
28°0'0"N
27°0'0"N
N
W
E
S
Khanuwala
Chhatargarh
Lunkaransar
Poogal
Bajju
Bikaner
Sridungargarh
Kolayat
Jasrasar
Hadan
Nokha
Bikaner
High : 536.31
Low : 310.59
0 15 30 60 Km


Mean Annual Rainfall 2020-2024 by Subdistrict
Mean Annual Rainfall
Mean Annual Rainfall (mm)
500
450
400
350
300
250
200
Bajju
Bikaner
Chhatargarh
Hadan
Jasrasar
Khajuwala
Kolayat
Lunkaransar
Nokha
Poogal
Sridungargarh
Subdistrict

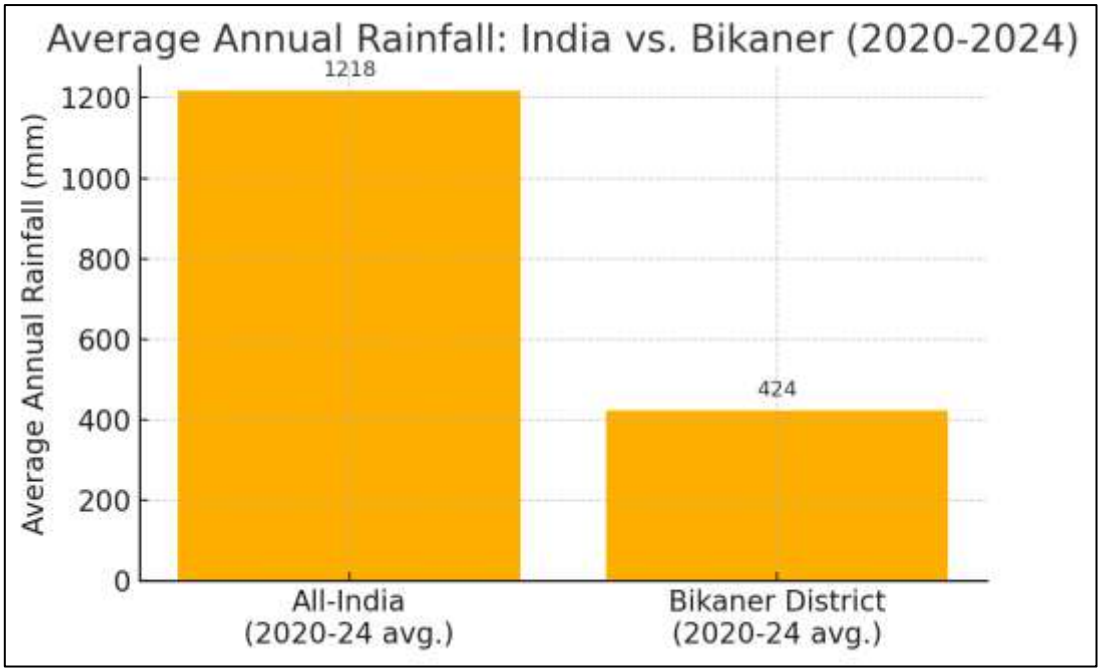
To compare the average annual rainfall of Bikaner District (2020–2024) with national figures:

- Bikaner District (as per map):
  Rainfall ranges from 310.59 mm to 536.31 mm annually.
- India's All-India Average Annual Rainfall (based on IMD data):
  - 2020: ~1094 mm
  - 2021: ~1054 mm
  - 2022: ~1087 mm
  - 2023: ~1042 mm
  - 2024: Preliminary estimates suggest similar range (~1050–1100 mm)

Comparison Summary:

- Bikaner receives significantly less rainfall than the all-India average — roughly half or even less in some areas.
- This indicates a semi-arid climate with high vulnerability to droughts and limited water availability, especially when coupled with rising temperatures and frequent heat waves.

➢ Here's a quick visual that puts Bikaner's rainfall in context against the national picture for the 2020-2024 period:



Statistic (2020-24 average)	All-India	Bikaner District
Average annual rainfall	**≈ 1219 mm**	**≈ 424 mm**

- Bikaner receives **only about one-third of the national mean rainfall**.
- The gap illustrates why Bikaner's heat-stress and water-scarcity risks are so acute: limited precipitation means less soil-moisture buffering and fewer opportunities to recover between increasingly frequent heatwaves.

Sources (national totals) – IMD "Rainfall Statistics of India" annual reports: 2020 (1 289.6 mm) (Hydro IMD), 2021 (1 236.4 mm) (Hydro IMD), 2022 (1 257.0 mm) (Hydro IMD), 2023 (1 102.8 mm) (metnet.imd.gov.in), 2024 (1 206.6 mm, provisional) (Hydro IMD).

Source (district range) – Your 2020-24 Bikaner rainfall map (low ≈ 310 mm; high ≈ 536 mm). The plotted value (~424 mm) is the mid-range average.

Take-away: Bikaner's climate is not just hotter; it is persistently drier than the country. Any heat-wave adaptation strategy must therefore integrate aggressive water-resource management (rainwater harvesting, micro-irrigation, drought-resilient crops) alongside public-health heat-action plans.

Conclusion

The analysis reveals a significant disparity between the rainfall patterns of Bikaner district and the national average. While the all-India average annual rainfall between 2020 and 2024 ranged between 1040 mm to 1100 mm, Bikaner district received substantially lower rainfall, ranging from just 310.59 mm to 536.31 mm. This highlights the region's semi-arid nature and increased susceptibility to water scarcity. Coupled with the observed rise in average maximum temperatures and the growing frequency, intensity, and duration of heat waves, the region faces escalating climate stress. The limited rainfall further compounds the vulnerability of the population by reducing agricultural productivity and increasing dependence on limited water resources. These findings underscore the urgent need for localized heat action plans, improved water management, drought-resilient cropping strategies, and community-based climate adaptation measures to ensure long-term resilience and sustainable livelihoods in Bikaner district.